

SKILL-01

To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

CODE:-

```
child@kali:~/lab$ cat skill-01.c
Child process completed.
root@kali:~/lab$ cat skill-01.c
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child Process
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is executing 'ls -l' using exec()\n\n");

        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        exit(1);
    }
    else
    {
        // Parent Process
```

```

printf("Parent Process Started\n");
printf("Parent PID : %d\n", getpid());
printf("Parent PPID : %d\n\n", getppid());

pid = fork();

if (pid < 0)
{
    perror("fork failed");
    return 1;
}

if (pid == 0)
{
    // Child Process
    printf("Child Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Child PPID : %d\n\n", getppid());

    printf("Child is executing 'ls -l' using exec()\n\n");

    execlp("ls", "ls", "-l", NULL);

    perror("exec failed");
    exit(1);
}
else
{
    // Parent Process
    printf("Parent Process\n");
    printf("Child PID : %d\n", pid);

    wait(NULL);

    printf("\nChild process completed.\n");
}

return 0;
}
$ make BUA1TOP-KHLAB07F:~/os-lab/skill1$

```

OUTPUT:-

```
root@APT0P-MHRLR30TF:~/os-lab/skill$ nano skill-01.c
root@APT0P-MHRLR30TF:~/os-lab/skill$ gcc skill-01.c -o skill.c
root@APT0P-MHRLR30TF:~/os-lab/skill$ ./skill-01.c
-bash: ./skill-01.c: Permission denied
root@APT0P-MHRLR30TF:~/os-lab/skill$ ls -l
total 36
-rwxr-xr-x 1 root root 16296 Aug 17 04:51 skill-01
-rw-r--r-- 1 root root 916 Aug 17 04:57 skill-01.c
-rwxr-xr-x 1 root root 16296 Aug 17 04:57 skill.c
root@APT0P-MHRLR30TF:~/os-lab/skill$ gcc skill-01.c -o skill-01
root@APT0P-MHRLR30TF:~/os-lab/skill$ ./skill-01
Parent Process Started
Parent PID : 925
Parent PPID : 437

Parent Process
Child PID : 926
Child Process
Child PID : 926
Child PPID : 925

Child is executing 'ls -l' using exec()

total 36
-rwxr-xr-x 1 root root 16296 Aug 17 04:58 skill-01
-rw-r--r-- 1 root root 916 Aug 17 04:57 skill-01.c
-rwxr-xr-x 1 root root 16296 Aug 17 04:57 skill.c

Child process completed.
```